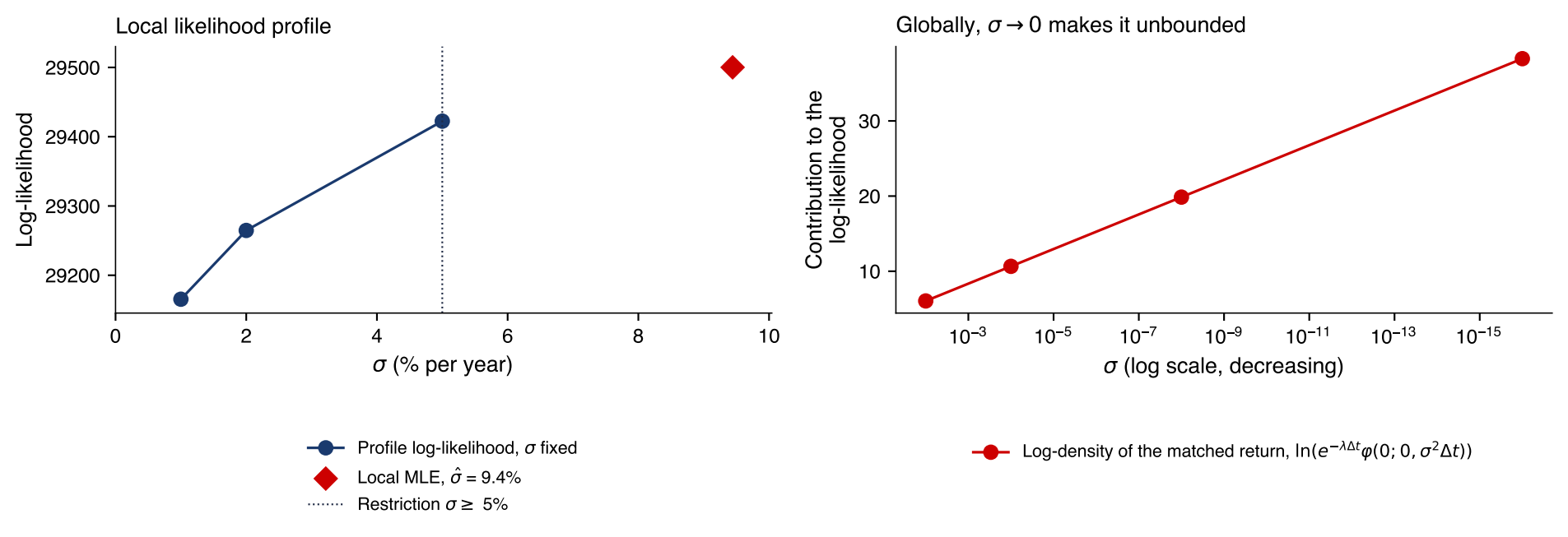
Local likelihood profile
Log-likelihood
29500
29400
29300
29200
0
2
4
6
8
10
σ (% per year)
Profile log-likelihood, σ fixed
Local MLE, σ̂ = 9.4%
Restriction σ ≥ 5%
Globally, σ → 0 makes it unbounded
Contribution to the log-likelihood
30
20
10
10^−3
10^−5
10^−7
10^−9
10^−11
10^−13
10^−15
σ (log scale, decreasing)
Log-density of the matched return, ln(e^−λΔt φ(0; 0, σ²Δt))